

Lab 1:

Session A:

```
1 begin;
2
3 update film set rental_rate = 4.99 where film_id = 1;
4 select film_id, rental_rate from film where film_id = 1;
```

Data Output Messages Notifications

	film_id [PK] integer	rental_rate numeric (4,2)
1	1	4.99

Session B:

```
5
6 begin;
7
8 update film set rental_rate = 5.99 where film_id = 1;
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
UPDATE 1

Query returned successfully in 171 msec.

Chờ session A nhả lock

- Quay lại session A commit:

Data Output Messages Notifications

COMMIT

Query returned successfully in 91 msec.

Sau khi commit session thì chạy lại đã thấy cập nhật lại kq ở session B:

Data Output			Messages	Notifications
	film_id	rental_rate		
	[PK] integer	numeric (4,2)		
1	1	5.99		

Lab 2:

Session A:

--step 1

11	lab 2
12	begin;
13	
14	select rental_rate from film where film_id = 10;
15	
16	
17	

Data Output			Messages	Notifications
	rental_rate			
	numeric (4,2)			
1	4.99			

Session B:

--step 2

```

16 begin;
17
18 update film set rental_rate = 0.99 where film_id = 10;
19
20 commit;

```

Data Output Messages Notifications

WARNING: there is already a transaction in progress
COMMIT

Query returned successfully in 47 msec.

Step 3: update thanh cong vi commit roi

```

22 select rental_rate from film where film_id = 10;
23
24
25

```

Data Output Messages Notifications

	rental_rate numeric (4,2) 
1	0.99

Lab 3:

Session A:

--step 1

```

23
24 -- lab 3
25 begin;
26
27 update film set rental_rate = 2.99 where film_id = 100;

```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 48 msec.

Sau câu lệnh này session A đang giữ **X-lock** trên row

--step 2: Session B khóa film_id = 200

```
28  
29 begin;  
30  
31 update film set rental_rate = 2.99 where film_id = 200;  
32
```

Data Output Messages Notifications

WARNING: there is already a transaction in progress

UPDATE 1

Query returned successfully in 42 msec.

--step 3: Session A muốn lấy row 200

```
5 update film set rental_rate = 3.99 where film_id = 200;
```

Data Output Messages Notifications

Bị treo vì B đang giữ row 200

Đợi rollback

--step 4: Session B muốn lấy row 100

```
5 update film set rental_rate = 3.99 where film_id = 100;
```

Data Output Messages Notifications

ERROR: deadlock detected
Process 34820 waits for ShareLock on transaction 992; blocked by process 24992.
Process 24992 waits for ShareLock on transaction 993; blocked by process 34820.

SQL state: 40P01
Detail: Process 34820 waits for ShareLock on transaction 992; blocked by process 24992.
Process 24992 waits for ShareLock on transaction 993; blocked by process 34820.
Hint: See server log for query details.
Context: while locking tuple (5,12) in relation "film"

Bây giờ postgresSQL phát hiện A chờ B, B chờ A => deadlock

Lab 4:

Step 1: Session A khóa DVD

```
8
9  begin;
10
11  select *
12  from inventory
13  where inventory_id = 10 for update;
```

Data Output Messages Notifications

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17

Step 2: Session B muốn thuê cùng DVD

```
8
9  begin;
10
11  select *
12  from inventory
13  where inventory_id = 10 for update;
```

Data Output Messages Notifications

Lúc này query sẽ bị đứng vì nó đang chờ A nhả block

Step 3: Session A tạo rental

```

14
15 insert into rental
16 (rental_date, inventory_id, customer_id, staff_id)
17 values
18 (now(), 10, 1, 1);
19
20 commit;

```

Data Output **Messages** Notifications

COMMIT

Query returned successfully in 58 msec.

Bây giờ DVD đã được khóa an toàn.

Step 4: Session B tiếp tục

Sau khi A commit thì câu lúc step 2 sẽ chạy tiếp

Nhưng lúc này ứng dụng kiểm tra DVD đã có người thuê rồi.

Nên:

rollback;

Hủy transaction của Customer 2.

Data Output Messages Notifications

	inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2	2006-02-15 10:09:17